# PHYSICAL CHEMISTRY CUMULATIVE EXAM

## January 20, 1996

## *In electronic spectroscopy:*

1. What is the Franck-Condon principle?
(What do we mean by vertical transition, progression of vibration, the transition dipole moment, the Franck-Condon factor?)

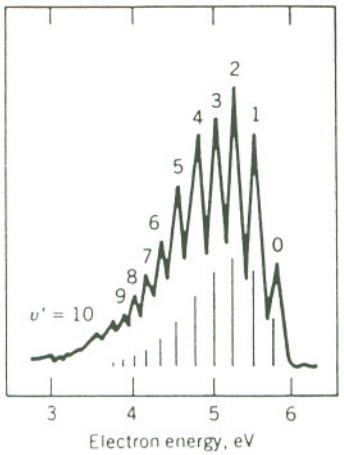
2. Describe the sequence of steps leading to fluorescence.

3. Describe the sequence of steps leading to phosphorescence.

4. Describe the steps involved in dissociation and predissociation.

5. What can be said about the vibrational structure observed in the absorption spectrum and in the fluorescence spectrum?

6. Describe the operations and principles involved in Photoelectron Spectroscopy.

7. The Figure below is the photoelectron spectrum of hydrogen gas reported by D. W. Turner and D. P. May, *J. Chem. Phys.*, **45** (1966) 471. Explain the spectrum and give an assignment of the observed lines. Explain their relative intensities.

(Use model molecular potential energy curves, where it is possible, to illustrate your answers.)